Euexia: A Gemma 4 Post-Consultation Care Companion

Turning doctor instructions into a personalized recovery map with structured tasks, grounded follow-up chat, and gamified adherence support.

Summary

Euexia is a mobile-first post-consultation care companion for patients who have just finished a doctor visit. The core problem is simple but serious: patients leave consultations with medication instructions, lab tests, follow-up appointments, lifestyle advice, and warning signs, but much of that plan is forgotten, misunderstood, or never converted into daily action.

Euexia uses Gemma 4 on Google Vertex AI to transform visit notes, PDFs, and medical document images into a structured recovery journey. The app generates scheduled checklist tasks, maps them into day-based game checkpoints, rewards progress with XP and coins, and provides a contextual “Dr. Gemma” assistant grounded in the patient’s saved consultation.

GitHub: https://github.com/elbochh/euexia
Live Demo: https://euexia-frontend-hgltl3cogq-uc.a.run.app/

Problem Statement

The failure point after a consultation is not only medical knowledge; it is continuity. Patients are often expected to remember complex care plans and execute them alone.

A review in the Journal of the Royal Society of Medicine found that 40-80% of medical information is forgotten immediately, and nearly half of remembered information is recalled incorrectly [1]. CDC reports that 1 in 5 new prescriptions are never filled, and about 50% of filled prescriptions are taken incorrectly, especially around timing, dosage, frequency, or duration [2]. U.S. health literacy guidance also notes that nearly 9 in 10 adults have limited health literacy skills [3]. AHRQ further reports that patients with clear post-discharge understanding are 30% less likely to be readmitted or visit the emergency department [4].

This means the post-consultation gap is not caused by patients “not caring.” It is caused by information overload, stress, low recall, complex regimens, unclear follow-up systems, and weak reinforcement after the visit. Euexia targets this exact moment: after the doctor has given the plan, but before the patient loses the thread.

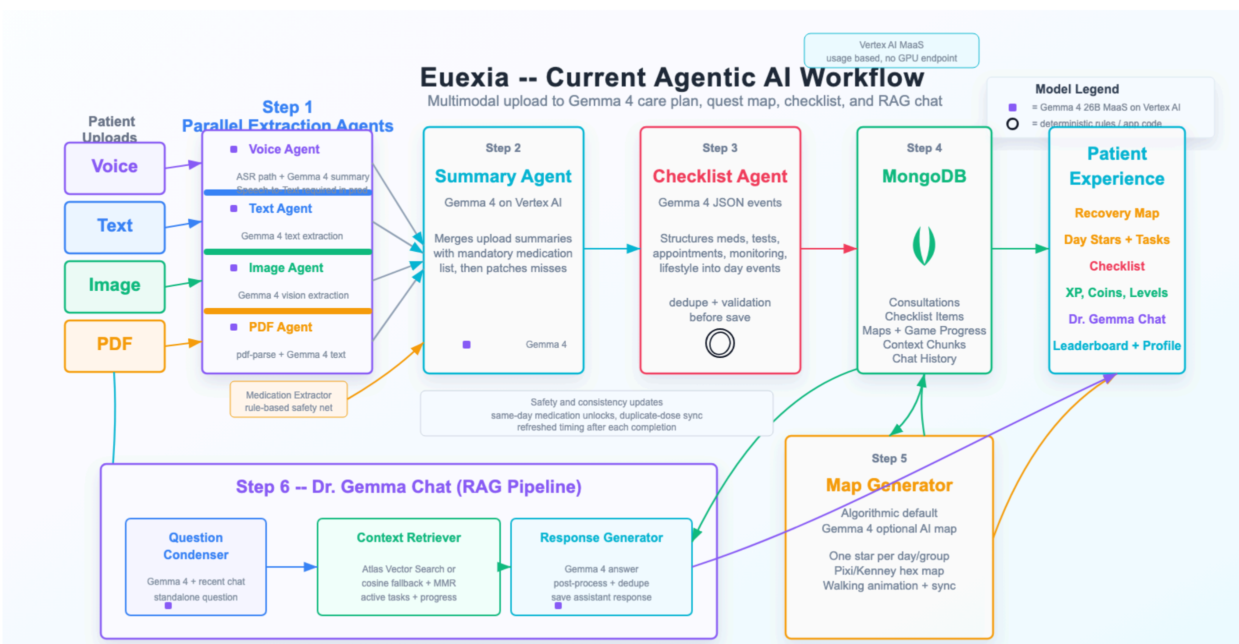
Approach

Euexia turns static consultation information into a living care journey.

Patients can upload consultation notes as text, images, or PDFs. Gemma 4 extracts key medical details including medications, doses, frequencies, durations, lab work, follow-up appointments, monitoring instructions, lifestyle advice, and warning signs. The system then creates scheduled checklist events and stores them in MongoDB.

The patient experience is intentionally mobile-first. Instead of reading a long discharge document, the patient sees a recovery map. Each day has checkpoints. Each checkpoint contains concrete actions such as taking a medication, booking a follow-up, checking INR, repeating blood work, walking, hydrating, or monitoring symptoms.

Agentic Workflow



The backend uses a modular agentic pipeline:

1. Parallel extraction agents process text, image, and PDF inputs.
2. A summary agent merges all extracted information into a unified care plan.
3. A rule-based medication extractor acts as a safety net so medication names are not silently missed.
4. A checklist agent converts the care plan into scheduled JSON events.
5. Validators repair malformed JSON, deduplicate tasks, normalize categories, and check medication completeness.
6. A map generator converts task groups into day-based game checkpoints.
7. A RAG pipeline powers Dr. Gemma chat by retrieving the user's consultation summary, checklist items, and recent chat context.

This hybrid design matters. Gemma 4 provides flexible multimodal reasoning, while deterministic validators protect safety-critical details like medication completeness.

Why Gemma 4?

Gemma 4 is a strong fit because Euexia is not a generic chatbot. It requires multimodal clinical document understanding, structured JSON generation, long-context care-plan extraction, and grounded follow-up support.

Gemma 4 also fits the deployment goal: it is an open model family available through Vertex AI Model-as-a-Service, which lets us build without managing GPU inference infrastructure while still aligning with privacy-sensitive and potentially local-first healthcare use cases.

We evaluated Gemma 4 against GPT-5.4 mini on the same UH Bristol discharge summary sample. Each model ran the full workflow 5 times: PDF extraction, summary generation, care-plan generation, checklist JSON generation, parsing, and scoring.

Metric	Gemma 4	GPT-5.4 mini
Completed runs	5/5	5/5
Valid JSON	5/5	5/5
Medication-perfect runs	5/5	5/5
Checklist-quality pass	4/5	0/5
Average workflow time	20.9s	47.2s
Average estimated cost	$0.00284	$0.04290
Average total tokens	7,137	12,308

Both models captured all 10 expected medications in every run. However, GPT-5.4 mini repeatedly expanded medications into many duplicate checklist events, producing 39-69 events per run. Gemma 4 produced more compact, app-ready outputs with 16-22 events and usually no duplicate titles. For Euexia, this matters more than raw verbosity: patients need clear daily actions, not noisy task overload.

The result supports our model choice: Gemma 4 delivered the medication-critical workflow with comparable recall, better checklist usability, faster latency, and much lower estimated cost.

Why Gamification?

Medical adherence is repetitive. Many tasks, such as taking medication, checking symptoms, booking follow-ups, or exercising, do not provide immediate reward. Euexia uses gamification to make invisible recovery progress visible.

The app uses map progression, XP, coins, streaks, character customization, and leaderboards. These mechanics are not decoration; they support behavioral follow-through. Previous research shows real-world health apps often suffer steep abandonment, including very low 30-day retention in mental health apps and high early abandonment in lifestyle and behavior-change apps [5][6]. Euexia responds by converting care into small, repeatable progress loops: complete a real task, unlock the next checkpoint, earn feedback, and continue the route.

Impact

Euexia is designed for the Health & Sciences track because it addresses a common real-world healthcare gap: what happens after the consultation. It does not replace clinicians, diagnose conditions, or invent treatment plans. Instead, it helps patients understand and follow the plan their doctor already gave them.

The long-term vision is a post-consultation companion that can be launched by QR code before a patient leaves the clinic, transforming discharge papers and visit notes into an accessible mobile journey.

Current Limitations

Euexia is a proof-of-concept. It requires further clinical validation, stronger safety review, and integration with provider workflows before real medical deployment. Voice transcription is part of the architecture but should use a production Google Speech-to-Text path before being relied on. All AI outputs should remain supportive, not authoritative medical advice.

References

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